Dataset (part 1 and 2)

The Bedtime Addiction: Sleep Debt & Screen Time dataset comprises 8,500 observations and 18 variables, including one user ID. Each observation represents an individual, with data covering demographic characteristics, bedtime technology habits, sleep patterns and lifestyle indicators. The dataset was synthetically generated using Python and calibrated against sleep research benchmarks. Although the data do not represent actual individuals, they provide a structured basis for exploring potential relationships and identifying patterns between bedtime habits and sleep outcomes before conducting real-world studies.

This analysis focuses on six variables: age, occupation type, chronotype, bedtime phone minutes, total sleep hours and next-day fatigue score. These variables were selected based on their relevance to the study of bedtime phone usage and its potential relationship with sleep outcomes. Age and occupation provide demographic context, while chronotype captures the respondent's natural sleep - wake preferences, allowing comparisons between morning-oriented individuals, night owls and those in between. Bedtime phone minutes and total sleep hours are used to examine the relationship between screen usage and sleep duration, while next-day fatigue score provides an additional measure of the potential impact on individuals' daytime well-being. Of the six variables, four are stored as numeric in R (age, bedtime phone minutes, total sleep hours and next-day fatigue score), while occupation type and chronotype are stored as character variables and treated as categorical data.

Short summaries

On average, the 8,500 participants slept 6.27 hours (median 6.33, SD 1.45, range 3.2–9.8 hours) and used their phones for 59.2 minutes before bed (median 51, SD 36.9, range from 1-180 minutes). For phone use, the mean is higher than the median, which suggests a small group of heavy users pulls the average up. Grouped by chronotype, bedtime phone use is almost identical (58.3–59.8 minutes), but sleep differs sharply. Morning Larks have 7.38 hours of sleep on average, Intermediates sleep around 6.44 hours, and Night Owls only have 5.01 hours. Night Owls also have the lowest maximum (8.17 hours) and the smallest spread (5 hours), which suggests a consistent short sleep pattern. After adding the variable of occupation, it shows that Morning Larks have the longest sleep duration on average, Intermediates are in the middle, and Night Owls have the least, still holding the power even across different jobs. These summaries already illustrate the pattern and potential relationship between these variables, through visualization, the relationship will become clearer and more straightforward.

Visualisation (part 3)

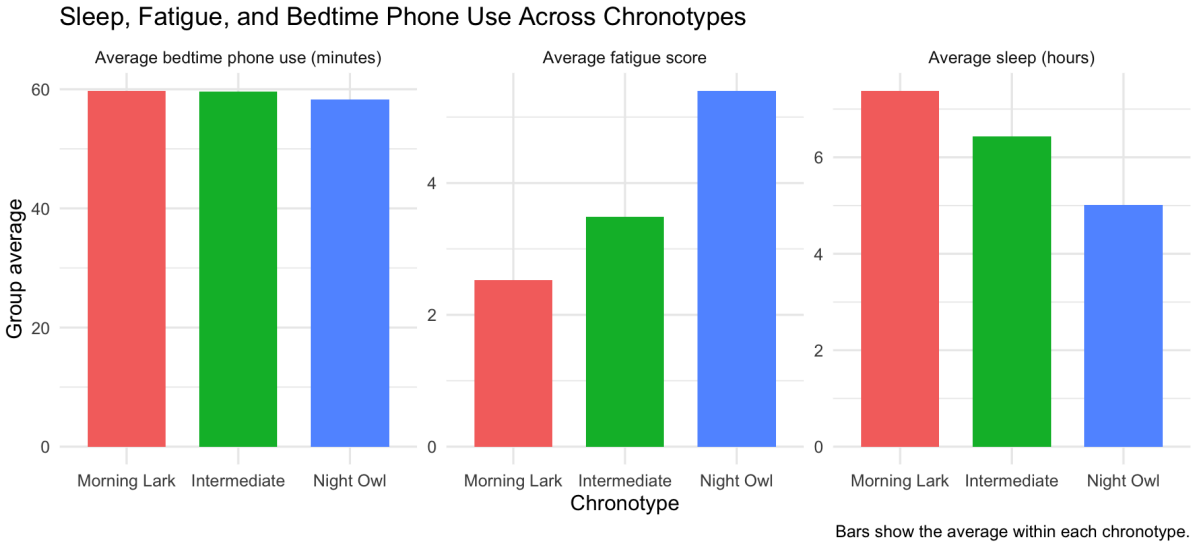


Figure 1: Faceted bar chart of group averages group by chronotype

Figure 1 compares bedtime phone use, fatigue score, sleep duration across Morning Larks, Intermediate chronotypes, and Night Owls. Separate panels show each group's average bedtime phone use(minutes), fatigue score, and sleep duration. Since these measures use different units and scales, the faceted layout makes it easier to compare chronotypes within each measure.

The chart suggests that Morning Larks have the highest average sleep duration and the lowest average fatigue score. Night Owls show the opposite pattern, while Intermediate chronotypes tend to fall between the two groups. Average bedtime phone use appears fairly similar across the three groups. This preliminary pattern suggests that chronotype may be associated with group differences in sleep duration and fatigue. However, averages do not show variation among individuals, and this chart does not establish whether the relationship between phone use and sleep differs by chronotype.

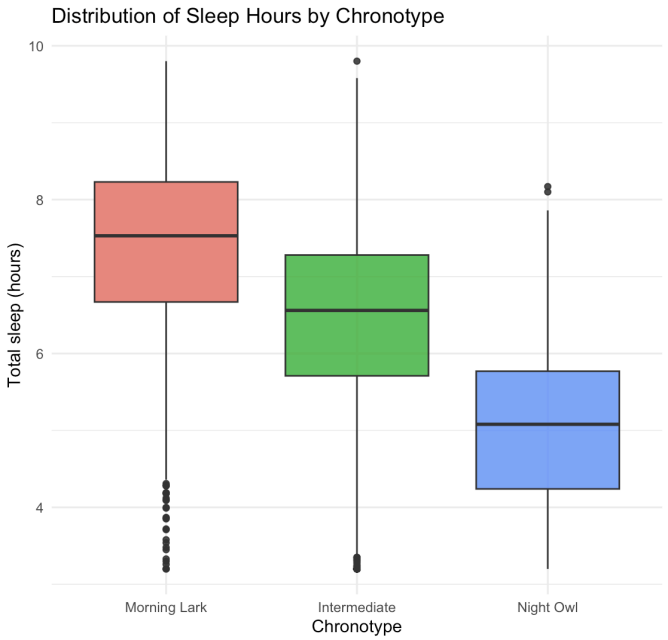


Figure 2: Distribution of sleep hours by chronotype

Figure 2 shows a significant gap in sleep hours by chronotype. Specifically, 50% of Morning Larks sleep between 6.7 and 8.2 hours, higher than Night Owls, which all fall below 6 hours. These two boxes do not overlap at all, indicating that a typical Night Owl sleeps less than almost any typical Morning Lark. Moreover, Intermediates sit between them (around 5.7–7.2 hours). For Morning Larks and Intermediates, short sleep below 4.3 hours appears as outliers, however, it is within the normal range of Night Owls, indicating that short sleep is common for this group of people.

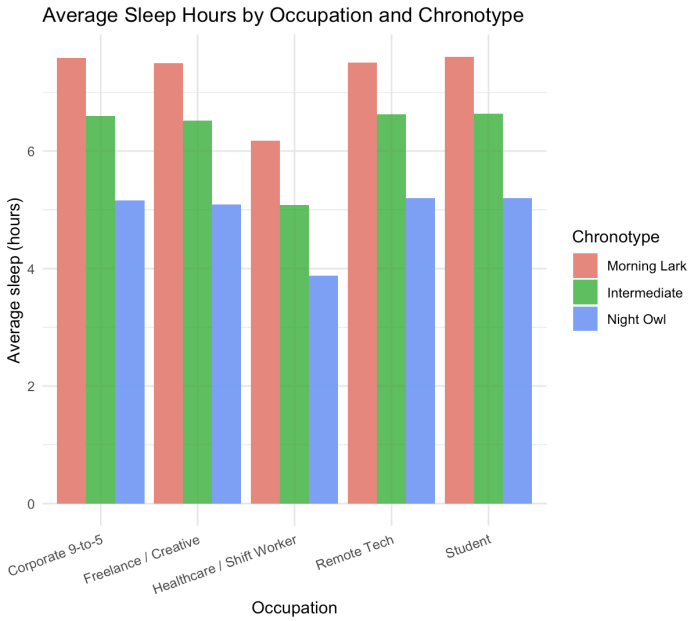


Figure 3: Average sleep hours by occupation and chronotype

Similar to the summary, Figure 3 shows the same pattern of average sleep hours on different chronotypes appears in all five occupations (Morning Larks the longest, followed by Intermediate and Night Owls). Most groups show similar sleep levels, such as the groups of Corporate, freelance, remote tech, and student. One exception is the group of Healthcare and shift workers, who sleep about 1 hour less than other occupations overall. Moreover, Night Owl shift workers have the lowest average of any subgroup. This indicates that occupation lowers sleep durations overall, but does not change chronotype patterns. It also shows one limitation of the simulated data, which is less dynamic and varied than real situations will be, since it is unlikely all four types of occupations have almost similar sleep levels in reality. However, it still gives us insights into special occupations, such as healthcare and shift workers.

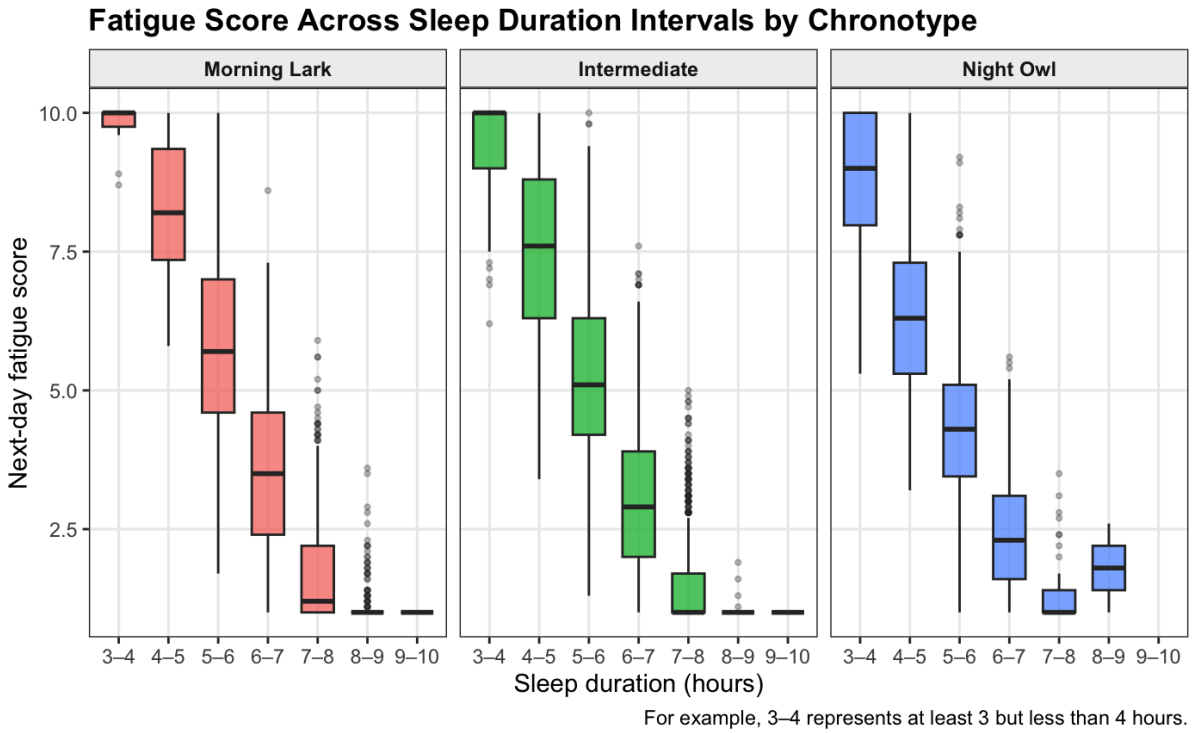


Figure 4: Next day fatigue scores across sleep duration groups by chronotype

Figure 4 shows a negative association between sleep duration and next day fatigue scores across all three chronotypes. Median fatigue scores of each interval generally decrease as sleep duration increases from approximately three to eight hours. However, the spread of scores within each interval shows that individuals with similar sleep durations can still experience different levels of fatigue.

Although the earlier faceted bar chart (Figure 1) shows higher average fatigue among Night Owls, this pattern is less obvious when comparing individuals within the same sleep duration interval. Night Owls often have similar or lower median fatigue scores than the other chronotypes within these intervals. This suggests that differences in sleep duration may help explain the overall fatigue differences between chronotypes.

For longer sleep durations, many fatigue scores cluster near the lower limit, producing very narrow boxes. The slight increase among Night Owls in the 8-9 hours interval contains only two observations, which contribute little to alter the overall pattern of decreasing fatigue with longer sleep duration.

Modeling (part 4)

Motivation and Comparability

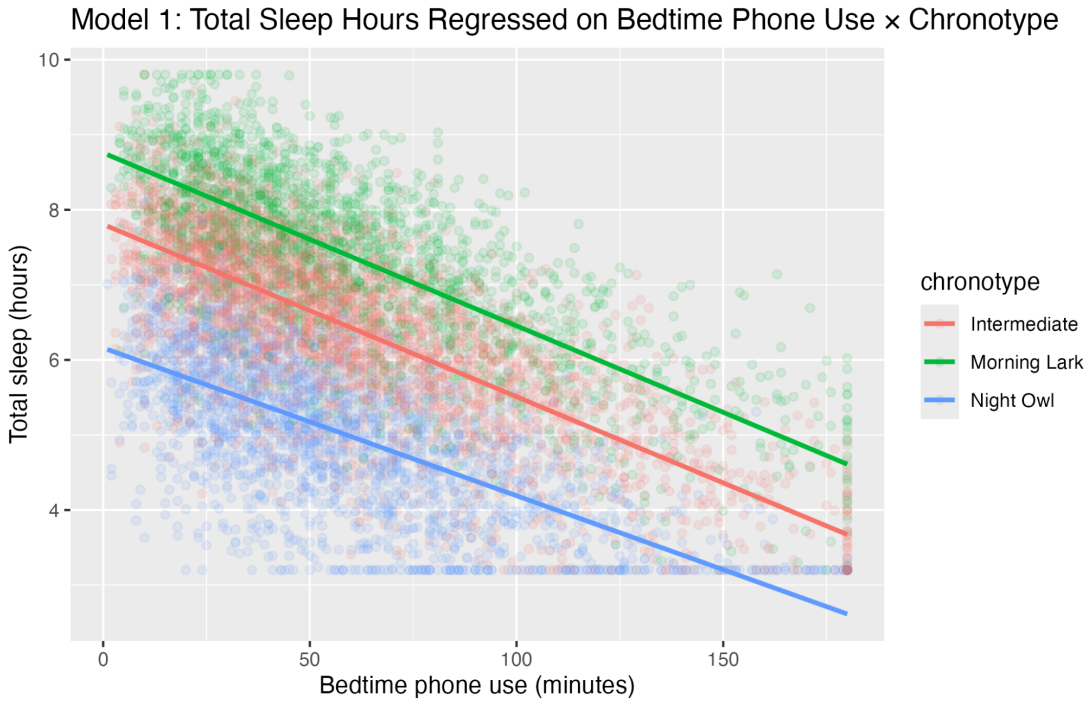


Figure 5 Regression Model

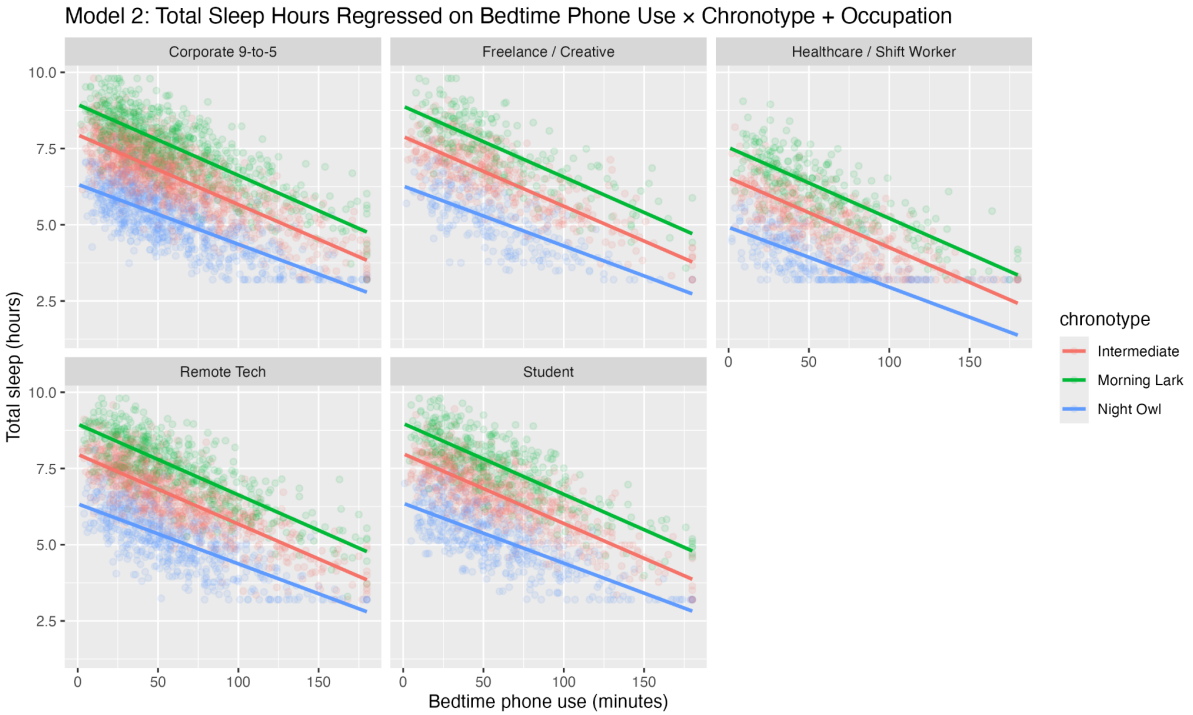


Figure 6 Interaction Model

Model 1 is a linear regression model with total sleep hours as the dependent variable. It includes the independent variables bedtime phone minutes, chronotype and the interaction between bedtime phone minutes and chronotype. Our formal findings showed a negative correlation between phone use and sleep. To investigate whether or not this effect varies across chronotypes, we created Model 1. An interaction model was selected instead of an additive model. The interaction term gives each chronotype its own slope since phone use may have more of an influence on some chronotypes than others.

In Model 2, occupation is added to the first model. It has the same dependent variable (total sleep hours) as Model 1. Its independent variables are bedtime phone minutes, chronotype, the interaction of bedtime phone minutes and chronotype, and occupation. We added occupation due to a finding in Model 1. A similar effect of phone use on sleep was found in all three chronotypes. The main difference between chronotypes was the quantity of sleep they had. This prompted us to wonder what other aspects could be the reason behind some people sleeping less than others. Figure 3 pointed to occupation as one answer. Healthcare / Shift Workers slept less than other occupations in all three chronotypes. However, Figure 3 only compared group averages. Therefore, we created Model 2 in order to see if sleep is explained by occupation, over and above phone use and chronotype.

It is possible to compare Model 2 with Model 1 since they have the same dependent variable. They differ only in their independent variables. Model 2 has only one new variable that it adds to Model 1. Thus Model 1 is nested within Model 2. Any difference between the two models is due only to occupation. The comparison illustrates whether occupation explains additional variation in sleep above and beyond phone use and chronotype.

Model 2 is the better model. Since both models have the same dependent variable and Model 2 only adds occupation to Model 1, they can be compared directly using adjusted R^2. Adjusted R^2 measures how much of the variation in sleep a model explains, while penalising unnecessary variables. Model 2 has a higher adjusted R^2 (0.794 vs 0.695), meaning it explains about 79% of the variation in sleep compared with about 69% for Model 1. This increase of roughly 10 percentage points shows that adding occupation helps explain differences in sleep, even after accounting for the extra variable. Therefore, Model 2 is preferred.

In both models, the more time people spend on their phone before bed, the less they sleep. In Model 1, each extra hour of bedtime phone use is associated with about 1.3 hours less sleep. The three chronotype lines are close to parallel, which means phone use affects Morning Larks, Intermediates and Night Owls in a similar way. In other words, the effect of phone use does not depend much on chronotype. What differs between chronotypes is the starting level of sleep. Compared with Morning Larks, Intermediate types sleep about 1 hour less and Night Owls about 2.5 hours less, at any given level of phone use. For example, a Night Owl using their phone for 3 hours before bed is predicted to sleep only about 2.5 hours.

Model 2 shows the same pattern for phone use and chronotype, so these two findings are not simply caused by occupation. Occupation adds one further finding: Healthcare / Shift Workers sleep about 1.4 hours less than people in the other occupations, whatever their

chronotype or phone use. Corporate 9-to-5, Freelance / Creative, Remote Tech and Student groups are very similar to one another. Adding occupation raises the share of sleep variation explained from about 69% to 79% (adjusted R^2 of 0.695 vs 0.794), and most of this improvement appears to come from the lower sleep of Healthcare / Shift Workers. The worst-off group is therefore Night Owls who work in healthcare or shifts and use their phone heavily before bed.

These results show that people who already sleep badly may use their phones more, rather than the phone causing the lost sleep. Also, the data appear to have a minimum of about 3.2 hours, with many points stacked at that floor, so the straight lines slightly overstate how low sleep falls for the heaviest phone users.